

HANNAH KIM, PH.D., M.S.

Temple University ◇ 1925 N 12th St, Room 643, Philadelphia, PA 19122

hannah.kim0007@temple.edu ◇ hk.garnet@gmail.com

website: hannahkimincompbio.github.io

WHO AM I?

- A passionate researcher seeking a post-doctoral position starting in Fall 2026.
- A community builder who is good at initiating and facilitating collaborations.
- An academic hybrid with a strong desire to foster [an inclusive interdisciplinary learning environment](#).
- Research interests in bioinformatics, human-computer interaction, and interdisciplinary learning.

EDUCATION

PhD in Bioinformatics, Temple University 2019 - 2026

Advisor, Bioinformatics: Dr. Sergei L Kosakovsky Pond

Co-Advisor, Human-Computer Interaction (HCI): Dr. Stephen MacNeil

Dissertation Topic: Harnessing interdisciplinarity in the investigation of nucleotide sequence evolution

[[Recap](#) | [Abstract](#) | [Shorts](#) | Dissertation]

MS in Computational Biology, Carnegie Mellon University 2015 - 2016

Relevant Coursework: Computational Genomics, Machine Learning, and Algorithms & Advanced Data Structures

BS in Chemistry, Carnegie Mellon University 2010 - 2013

Relevant Coursework: Principles of Computing, and Modern Analytical Instructions

MAJOR CERTIFICATES

08/2023 **Teaching in Higher Education Certificate** Temple University - Center for the Advancement of Teaching

08/2023 **MicroMBA** University of California, San Diego Extended Studies - Rady School of Management

PUBLICATION

[P4] **Kim, H.**, Scheffler, K., Nekrutenko, A., Martin, D. P., Weaver, S., Murrell, B., Kosakovsky Pond, S. L. (2026). Characterizing Physicochemical Selection in Protein Evolution with Property-Informed Models (PRIME). *Revision Submitted to Molecular Biology and Evolution*, 1-43. doi:[10.64898/2026.03.09.710461](https://doi.org/10.64898/2026.03.09.710461) [[Code](#) | [Web App](#) | [Data](#)]

[P3] **Kim, H.**, Nabid, R. A., Sorathiya, J., Doan, M., Jordan, E., Nasimova, R., Kosakovsky Pond, S. L., MacNeil, S. (2026). Changing the Optics: Comparing Traditional and Retrieval-Augmented GenAI E-Tutorials in Interdisciplinary Learning. *arXiv*, 1-16. doi:[10.48550/arXiv.2602.20544](https://doi.org/10.48550/arXiv.2602.20544) [[Post](#) | [Task](#) | [Trad E-Tutorial](#) | [RAG E-Tutorial](#)]

[C2] Auer, S., Betz, D., Binossek, C., Jacyszyn, A., Jiomekong, A., **Kim, H.**, McGinty, H.K., Mills, K.G., Oelen, A., Rabby, G. and Rajabi, E. (2026). Proceedings of the 2nd AAAI Bridge on Artificial Intelligence for Scholarly Communication. *Open Conference Proceedings*, 8. doi:[10.52825/ocp.v8i.3212](https://doi.org/10.52825/ocp.v8i.3212)

[P2] Jiomekong, A., McGinty, H. K., Mills, K. G., Oelen, A., Rajabi, E., McElroy, H., Christou, A., Saini, A., Zebaze, J. A., **Kim, H.**, Jacyszyn, A. M. (2025). Charting the future of scholarly knowledge with AI: A community perspective. *arXiv*, 1-39. doi:[10.48550/arXiv.2509.02581](https://doi.org/10.48550/arXiv.2509.02581)

[C1] **Kim, H.**, Kosakovsky Pond, S. L., MacNeil, S. (2025). Conversations over Clicks: Impact of Chatbots on Information Search in Interdisciplinary Learning. *2025 IEEE Frontiers in Education Conference (FIE)*, 1-9.

doi:[10.1109/FIE63693.2025.11328556](https://doi.org/10.1109/FIE63693.2025.11328556)

- [FIE 2025](#) [[Preprint](#) | [Teasers](#) | [Slide](#) | [Video](#) | [Recap](#)]

- [W2] **Kim, H.**, Kosakovsky Pond, S. L., MacNeil, S. (2024). WIP: Identifying Tutorial Affordances for Interdisciplinary Learning Environments. *2024 IEEE Frontiers in Education Conference (FIE)*, 1-5. doi:[10.1109/FIE61694.2024.10893187](https://doi.org/10.1109/FIE61694.2024.10893187)
- [FIE 2024](#) [[Preprint](#) | [Slide](#) | [Video](#) | [Recap](#)]
- [W1] **Kim, H.** (2023). Running a summer journal club for an interdisciplinary community: how to maintain engagement when members have disparate prior knowledge. *The Journal for Research and Practice in College Teaching*, 8(2). <https://journals.uc.edu/index.php/jrpct/article/view/8150/6710>
- [J7] Huzar, J., **Kim, H.**, Kumar, S., Miura, S. (2022). MOCA for integrated analysis of gene expression and genetic variation in single cells. *Frontiers in Genetics*, 13:831040. doi:[10.3389/fgene.2022.831040](https://doi.org/10.3389/fgene.2022.831040)
- [J6] Ding, Y., **Kim, H.**, Madden, K., Loftus, J., Chen, G., Allen, D., Zhang, R., Xu, J., Chen, C., Xu, Y., Tasian, S., Tan, K. (2021). Network Analysis Reveals Synergistic Genetic Dependencies for Rational Combination Therapy in Philadelphia Chromosome-like Acute Lymphoblastic Leukemia. *Clinical Cancer Research*, 27(18). doi:[10.1158/1078-0432.CCR-21-0553](https://doi.org/10.1158/1078-0432.CCR-21-0553)
- [J5] Tarca, A. L., Pataki, B. Á., Romero, R., Sirota, M., Guan, Y., Kutum, R., Gomez-Lopez, N., Done, B., Bhatti, G., Yu, T., Andreoletti, G., Chaiworapongsa, T., **The DREAM Preterm Birth Prediction Challenge Consortium**, Hassan, S. S., Hsu, C., Aghaeepour, N., Stolovitzky, G., Csabai, I., Costello, J. C. (2021). Crowdsourcing assessment of maternal blood multi-omics for predicting gestational age and preterm birth. *Cell Reports Medicine*, 2(6). doi:[10.1016/j.xcrm.2021.100323](https://doi.org/10.1016/j.xcrm.2021.100323)
- [J4] Ichikawa, Y., Bruno, V. M., Woolford, C. A., **Kim, H.**, Do, E., Brewer, G., Mitchell, A. P. (2021). Environmentally contingent control of *Candida albicans* cell wall integrity by transcriptional regulator Cup9. *Genetics*, 218(3). doi:[10.1093/genetics/iyab075](https://doi.org/10.1093/genetics/iyab075)
- [J3] Tao, Y., Rajaraman, A., Cui, X., Cui, Z., Chen, H., Zhao, Y., Eaton, J., **Kim, H.**, Ma, J., Schwartz, R. (2021). Assessing the Contribution of Tumor Mutational Phenotypes to Cancer Progression Risk. *PLOS Computational Biology*, 17(3). doi:[10.1371/journal.pcbi.1008777](https://doi.org/10.1371/journal.pcbi.1008777)
- [J2] He, B., Gao, P., Ding, Y., Chen, C., Chen, G., Chen, C., **Kim, H.**, Tasian, S. K., Hunger, S. P., Tan, K. (2020). Diverse noncoding mutations contribute to deregulation of cis-regulatory landscape in pediatric cancers. *Science Advances*, 6(30). doi:[10.1126/sciadv.aba3064](https://doi.org/10.1126/sciadv.aba3064)
- [P1] Tao, Y., Rajaraman, A., Cui, X., Cui, Z., Eaton, J., **Kim, H.**, Ma, J., Schwartz, R. (2019). Improving personalized prediction of cancer prognoses with clonal evolution models. *bioRxiv*. doi:[10.1101/761510](https://doi.org/10.1101/761510)
- [J1] Lin, C., Jain, S., **Kim, H.**, Bar-Joseph, Z. (2017). Using neural networks for reducing the dimensions of single-cell RNA-Seq data. *Nucleic Acids Research*, 45(17). doi:[10.1093/nar/gkx681](https://doi.org/10.1093/nar/gkx681)

POSTERS

- Ahmed, M., Kashyap, D., **Kim, H.**, Duy, C.* (2025, April 25-30). *DNMT1 promotes escape from chemotherapy-induced senescence in acute myeloid leukemia* [Abstract]. In Proceedings of the 116th Annual Meeting of the American Association for Cancer Research. Chicago, IL: AACR. Abstract nr 0293 / 2.
- Kim, H.*** (2024). *It Takes a Village: Understanding the Community Aspects of Interdisciplinary Tutorials* Poster: 2024 ASEE/IEEE Frontiers in Education Conference, Washington, D.C.
- Kim, H.***, Kosakovsky Pond, S. L. (2023). *PRSuite: PProperty Informed Models of Evolution (PRIME), the Imputation (PREI), and the Visualization (PreC)*. Poster: The Society for Molecular Biology and Evolution 23 Conference, Ferrara, Italy.
- Kim, H.***, Shank, S., Kosakovsky Pond, S. L. (2022). *PProperty Informed Models of Evolution (PRIME)*. Poster: The 31st KSEA Northeast Regional Conference, Virtual.

Hu, Y., Chen, C., Ding, Y.*, **Kim, H.**, Tan, K. (2019). *Synergistic Control Genes in Cancer Gene Networks as Targets for Combination Therapy*. Poster: Children's Hospital of Philadelphia Research Poster day and Scientific Symposium, Philadelphia, PA.

INVITED TALKS

Kim, H.* (2026). *New Affordances and Limitations of E-Tutorials in the Era of Generative AI*. Invited Talk: Penn State Berks AI Week (April 14-17), Reading, PA. [[Flyer](#) | [Recap](#)]

Kim, H.* (2024). *Characterization of Disciplinary Hybridization in Bioinformatics Software Tutorials*. Seminar: Philadelphia Korean Scholars Association, Philadelphia, PA.

Kim, H.* (2023). *Using HyPhy package on comparative sequence analysis*. Workshop: Temple Bioinformatics Studio, Philadelphia, PA. [[Flyer](#)]

Kim, H.* (2023). *Data Analysis in Bioinformatics Research (in academia)*. Seminar: Philadelphia Developer Group, Virtual.

Kim, H.*, Kosakovsky Pond, S. L. (2022). *PRIME Evolutionary Imputation (PREI)*. Flash Talk: International Conference on Intelligent Biology and Medicine, Philadelphia, PA.

PROFESSIONAL EXPERIENCE

Project Lead Feb 2024 - May 2026
Temple University ACME Group & Temple University HCI lab Philadelphia, PA

- Identified gaps in knowledge and proposed ideas that can make data and method work together.
- Reconciled different research interests of stakeholders involved.
- Designed a user study, drafted a protocol approved by the institutional review board, and ran the study.
- Wrote a [\[P3\] Preprint about the strengths and limitations of GenAI when used as an e-tutorial](#).
- Team: Saumika Pandey (09/2025-04/2026 & 01/2025-05/2025); Hena Patel (05/2025-04/2026); Rayhana Nasimova (05/2025-04/2026); Elijah Jordan (05/2025-04/2026); Jeni Sorathiya (01/2025-04/2026); Rahad Arman Nabid (01/2025-04/2026); Kriti Baru (01/2025-05/2025); Minh Doan (01/2025-05/2025)

Bioinformatics Engineer Oct 2021 - Jan 2022
Lifetime Omics Remote

- Automated cutting-edge methods and analyzed COVID-19 metagenomics data in the Google Cloud environment.
- Developed detailed standard operating procedure for reproducibility.
- Took initiatives to solve problems and demonstrated dedication for the project in the startup environment.

Bioinformatics Analyst/Software Engineer Jul 2017 - Jun 2019
Children's Hospital of Philadelphia – Kai Tan lab Philadelphia, PA

- Investigated RNA-Seq and microarray data from B-Cell Acute Lymphoblastic Leukemia subtypes using differential gene expression analysis, Gene Ontology enrichment analysis, and other relevant bioinformatics methods.
- Identified cancer-specific genetic interactions that led to publications in high-impact journals.
- Facilitated communication in the interdisciplinary environment of doctors and wet-lab and dry-lab researchers.

Research Programmer Feb 2017 - Jun 2017
Carnegie Mellon University – Russell Schwartz lab Pittsburgh, PA

- Investigated clinical and genomic data to create a cancer progression analysis pipeline using machine learning.

MS Graduate Researcher Carnegie Mellon University – Ziv Bar-Joseph lab	Jan 2016 - Dec 2016 <i>Pittsburgh, PA</i>
Curated and analyzed single-cell gene expression data.	
Post-Baccalaureate Researcher (Chemistry/Biology) Carnegie Mellon University – Drs. Fred Lanni, Aaron Mitchell, and Luisa Hiller	Aug 2013 - Jun 2014 <i>Pittsburgh, PA</i>
Tested various polymer surface coatings for the prevention and destruction of biofilms using red/NIR light.	
Undergraduate Student Researcher (Chemistry) Carnegie Mellon University – Kevin Noonan lab	Jan 2012 - Aug 2013 <i>Pittsburgh, PA</i>
Characterized pyrylium and pyridinium salts by modifying functional groups.	
Student Researcher (Biology) Carnegie Mellon University – 2011 Summer Research Institute	Jun 2011 - Aug 2011 <i>Pittsburgh, PA</i>
- Analyzed interactions among three ribosomal assembly factors in <i>Saccharomyces cerevisiae</i>. 1 of 12 student researchers selected for the program.	

AWARDS, FELLOWSHIPS, & GRANTS

2023 Best Poster Award, 2nd Place (Computer and Information Sciences), The 36th US-Korea Conference

2022&2023 Selected Attendee Support, Scientists and Engineers Early Career Development Workshop

2022 KSEA Excellent Poster Award, The 31st KSEA Northeast Regional Conference

2022 CST Three-Minute Thesis Competition 2nd Place Award, Temple University (College-level)

2015 Departmental Merit Fellowship, Carnegie Mellon University

2013 Mellon College of Science Research Honors, Carnegie Mellon University

2012 Summer Undergraduate Research Fellowship, Carnegie Mellon University

TEACHING EXPERIENCE

Client Temple University	Jan 2025 - Apr 2025 <i>Projects In Computer Science (CIS-4398)</i>
- Provided feedback to a project as a client in Professor Ian Tyler Applebaum's senior capstone class for 6 weeks [Class Website Team Documentation Website Pitch as a Client Project Final Demo]. - Team size: 7.	
Co-Coordinator/Reflective Practicum Completion for THE Certificate Temple University	May 2023 - July 2023 <i>Deep Learning for the Life Sciences Journal Club</i>
- Co-organized a 10-week summer journal club with Dr. Enzo Carnevale for researchers of diverse levels of expertise (undergraduate students, graduate students, postdocs, alumni, visitors, and professors). - Built a unique inter-departmental community in the College of Science and Technology. - Discussed teaching methods and reflections with Dr. Jay Lunden (reflective practicum mentor) for 7 weeks. - Wrote a [W1] Personal Narrative of Works in Progress based on this experience. - Group size: 35.	
Course Completion for Teaching in Higher Education (THE) Certificate Temple University	Aug 2022 - Dec 2022 <i>Teaching in Higher Education (EPSY-8985)</i>

- Developed syllabi, assignments, and assessments using the principles of integrated course design.
- Applied the current theories of teaching in different contexts.
- Used a variety of effective teaching methods to address learners universally.
- Discussed a reflective and purposeful approach to teaching with other instructors.

Teaching Assistant

Temple University

Aug 2020 - Dec 2020

Genomics in Medicine (BIOL-3111/5111)

- Generated formative and summative assessment materials and provided timely feedback ([Youtube/Introduction](#)).
- Class size: 150.

Teaching Assistant

Temple University

Aug 2019 - May 2020

Wet Lab Courses (BIOL-2112 and BIOL-1012)

- Gave a short lecture in the beginning of every lab, monitored student performance, and provided guidance.
- Graded lab reports and generated quizzes.
- Class size: 20 (× 2 sections) each.
 - BIOL-1012 General Biology II, Spring 2020 was an introductory wet lab course for non-biology majors.
 - BIOL-2112 Introduction to Cellular and Molecular Biology, Fall 2019 was a lab for biology majors.

Course Developer

Carnegie Mellon University

Feb 2016 - Aug 2016

Programming for Scientists (02201/02601)

- Generated open-source course materials (codes and instructions) with Drs. Phillip Compeau and Carl Kingsford for Go-lang beginners.

TUTORIAL

[T2] Kim, H. [Hannah Kim in Comp Bio]. (2025, May 28). *#summer2025 #hci [Research Project Teaser] Bioinformatics Software Tutorial Supporting Chatbot* [Video]. YouTube. <https://youtu.be/cfTl6e58DPU>

[T1] Kim, H. [GalaxyProject]. (2024, Sep 13). *GTN Tutorial: Chloroplast Genome Assembly* [Video]. YouTube. <https://youtu.be/gIcbY9kXdTo?si=FrJLLb0aWNok-Ck>

UNIVERSITY SERVICE

Panelist

Oct 2024 - Aug 2025

- [CST Graduate Orientation, 08/2025](#), Graduate Admissions Team in College of Science and Technology
 - Shared graduate school tips and tricks with the new graduate students.
- [Publishing Journal Articles: A Panel Discussion, 10/2024](#), Temple University Graduate School
 - Served as one of four panelists. Shared my solo- and first-author experiences with graduate students across the community.

College of Science and Technology-Graduate Student Organization Board Member

Dec 2021 - May 2024

- **Media Chair**

May 2023 - May 2024

 - Advertised the organization events on social media. Communicated with members from different departments. Supported organization social events and generated flyers.
 - Organized Graduate Student Fall and Spring social events each with 30+ attendees with the board members.

• DEI Representative

Dec 2021 - Apr 2023

- Identified potential DEI topics within the organization and suggested appropriate strategies.
- Facilitated the success of the yearly PhD/MS alumni panelist event involving all six departments within college for two years in a row.

Student Representative

Nov 2021 - Aug 2023

Temple University College of Science and Technology Diversity, Equity, and Inclusion (CST-DEI) Committee

- Discussed the promotion of DEI within college at the bi-weekly committee meetings.
- Provided feedback for a wide array of DEI topics and addressed current issues with other members.

Vice President

Sep 2021 - Sep 2023

Temple University Biology Graduate Student Society (BGSS)

- Facilitated communication between the department and the graduate school.
- Evaluated travel grant applications with the president and the treasurer.
- Led action plans to address diverse career needs within the department of biology and organized regular meetings with the graduate school directors.
- Generated databases for student placements after graduation.
- Supported the on-boarding process for the first-year students in the department of biology.
- Planned and oversaw activities for the Fall 2022 departmental retreat.

LEADERSHIP EXPERIENCE

College of Science and Technology Alumni Board Member

June 2026 - present

[Temple University Alumni Association](#)

- Part of (1) Careers and Networking and (2) Events committees.

Education Standards Committee Member

Nov 2025 - present

[P3722 IEEE Standard Guide for Responsible AI in Higher Education](#)

- As a member of the working group P3722, participated in monthly meetings to discuss and advance the development of the standard.
- As a member of the subgroups (1) Overview and Guiding Principles, (2) Educator Development (specifically Institutional Support), and (3) External Awareness (specifically Risk Management), discussed the purpose and scope of the guideline, identified areas for further study, and drafted sections with future-proofing in mind.

Alumni Network Leader

Apr 2025 - present

[Carnegie Mellon University Philadelphia Network](#)

- Hosted local events ([Globally Plaid Recap, Sep 2025](#)) for the alumni and affiliates.

Bridge Organizing Committee Member

Sep 2025 - Jan 2026

[2nd AI for Scholarly Communication \(AI4SC\)](#) - the 40th Annual AAAI Conference on Artificial Intelligence ([AAAI-26](#))

- Contributed to the program document drafts, generated recruitment materials, and attended bi-weekly meetings.
- Co-managed bridge submissions as an editor on the [TIB Open Conference Proceedings](#) platform.

KITEE Leadership

Jan 2023 - present

• Young Generation Director I (PhD level)

Aug 2025 - present

- Supported organizing KITEE events and promoting membership drive.

- **The 33rd KSEA NRC Steering Committee Member & PR Director** Jan 2024 - Apr 2024
 - Planned the conference in the weekly committee meeting with PIs and staff from academia and industry.
 - Discussed conference planning with other members; an emphasis on diversity and inclusion (i.e. accessibility & childcare support).
 - Supported onsite facilitation on the day of event with >80 attendees.
- **Event Director** Oct 2023 - Dec 2024
 - Connected with graduate students in the northeast region in the US to promote KITEE events.
 - Brainstormed ideas during event meetings and supported all registration-related endeavors.
 - Organized [KITEE Entrepreneurship Forum 2023](#) (>80 attendees) as part of the committee.
- **The 2023 KITEE-FELIX Hackathon Committee Secretary** Feb 2023 - Apr 2023
 - Organized the hackathon in the weekly committee meeting.
 - Reached out to universities and engineer communities in the Philadelphia region.
 - Supported the onsite activities.
- **The 32nd KSEA NRC Committee Philadelphia Young Generation Director** Jan 2023 - Apr 2023
 - Planned the Northeast Regional Conference (NRC) conference in the bi-weekly committee meeting with PIs and staff from academia and industry.
 - Operated as the public relations and promotion chief for the pre-networking event before the conference.
 - Promoted networking (e.g. networking bingo) and facilitated the event flow onsite (104 registrants).

COMMUNITY INVOLVEMENT

Ad Hoc Reviewer

- KG4Health Workshop Program Committee as part of 24th International Conference on Artificial Intelligence in Medicine (AIME 2026), *05/2026*
 - Reviewed two papers about generative AI applications in health care.
- 2026 IEEE Frontiers in Education Conference (FIE) Technical Program Committee (TPC), *04/2026*
 - Reviewed two papers using background in informal learning.
- Virus Evolution (2024 IF=4.0), *01/2026*
 - Commented as a reviewer using knowledge in virus evolution and bioinformatics.
- 2026 IEEE Global Engineering Education Conference (EDUCON) TPC, *12/2025*
 - Reviewed two full papers and provided detailed feedback on machine learning models, including GenAI.
- Virus Evolution (2024 IF=4.0), *11/2025*
 - Commented as a reviewer drawing on the experience in evolutionary biology.
- 2025 IEEE FIE TPC, *04/2025*
 - Reviewed and provided feedback on two full papers, each with a strong focus on statistics and algorithms.
- 2025 IEEE EDUCON TPC, *11/2024*
 - Reviewed and provided feedback on three full papers about multidisciplinary and transdisciplinary education.
- 2024 IEEE FIE TPC, *07/2024 & 03/2024*
 - Reviewed and provided feedback on FIE conference abstracts (3) & papers (2).
- Temple University CST Diversity Innovation Initiative (DII) Fund, *03/2024*

- Scrutinized the funding proposals and provided feedback to the DII committee.
- [Scientific Reports \(2021 IF=4.996\)](#), 04/2023
 - Commented as a reviewer using the experience in cancer, evolutionary biology and single-cell transcriptomics.
- [Frontiers in Oncology \(2021 IF=5.738\)](#), 06/2022
 - Commented as a reviewer for an academic journal using my experience in cancer research and bioinformatics.

Science Fair Judge

- George Washington Carver Science Fair, 03/2026, 03/2025, 04/2024, 03/2023 & 03/2022
 - Judged science fair projects done by students in grades 4 and 5 (04/2024) and grades 6 to 12 (03/2026, 03/2025, 03/2022 & 03/2023) with a group of educators.
 - Judged Behavior and Social Sciences category in 03/2026 and 03/2025.

PROFESSIONAL MEMBERSHIPS

- [Knowledge Knocking for Your Universe \(KKYU\)](#), 05/2025-present
- [PhillyCHI](#), 01/2025-present
- [Galaxy Training Network Contributor](#), 07/2024-present
- [Temple Asian American and Pacific Islander \(AAPI\) Faculty & Staff Affinity Group](#), 02/2024-present
- [Korean-American Women in Science and Engineering \(KWise\)](#), 08/2023-present
- [Philadelphia Developer Group](#), 04/2023-present
- An Affiliate Member of Temple University [Human-Computer Interaction \(HCI\) Lab](#), 03/2023-present
- Collab with researchers from [Fox Chase Cancer Center](#) at Temple University Hosp., 12/2022-11/2024
- Collab with (current and previous) researchers from [Institute for Genomics and Evolutionary Medicine](#) at Temple University, 08/2022-11/2024
- [Korean American Society in Biotech and Pharmaceuticals \(KASBP\)](#), 10/2022-07/2024
- [Korean-American Innovative Technology Engineers and Entrepreneurs \(KITEE\)](#), 04/2022-present
- [AnitaB.org](#), 09/2021-present
- [Korean-American Scientists and Engineers Association \(KSEA\)](#), 02/2020-present
- Rotations in Zoran Obradovic, Sergei L. Kosakovsky Pond, and Sudhir Kumar Labs, 08/2019-05/2020
- [Philadelphia Korean Scholars Association \(PKSA\)](#), 06/2019-present

DOCTORAL COURSEWORK

Fall 2020 **Genomics and Infectious Disease Dynamics** *BIOL-5128*
 Fall 2020 **Seminar: "Ecoevo discuss"** *BIOL-8210*
 Fall 2020 **Applied Statistics and Data Science** *STAT-8109*
 Spring 2020 **Genomics and Evolutionary Biology of Parasites** *BIOL-5241*
 Spring 2020 **Data-Intensive and Cloud Computing** *CIS-5517*
 Spring 2020 **Knowledge Discovery and Data Mining** *CIS-5523*
 Fall 2019 **Genomics in Medicine** *BIOL-5111*
 Fall 2019 **Topics in Bioinformatics** *BIOL-5466*
 Fall 2019 **Seminar Biol 8210 at Center for Computational Genetics and Genomics** *BIOL-8210*

MASSIVE OPEN ONLINE COURSE CERTIFICATES

11/2022 **The Inclusive STEM Teaching Project** *InCLU1x*
 10/2022 **Business Foundations** *UBCx*
 09/2022 **Cancer Biology Specialization Courses** *Coursera*
 03/2022 **Matrix Algebra for Engineers** *Coursera*
 02/2022 **Viruses & How to Beat Them: Cells, Immunity, Vaccines** *IsraelX*

PERSONAL PROJECTS

HearU: Bridging the Gap in Korean-American Mental Health Care, 04/2023.

- Voted "Second Place" by the audience in the KITEE-FELIX Ideathon Pitch. Ideated by PhD students each from NJIT, Penn, and Temple.

Colorblindness Image Enhancer, 02/2023.

- "Most Technically Impressive" in OwlHacks 2023.

MEDIA APPEARANCES

The Aspiring STEM Geek Podcast, to appear in 07/2026.

- Invited to discuss my research experience at Children's Hospital of Philadelphia and Temple University. It was an enjoyable conversation about cancer genomics, evolutionary biology, human-computer interaction, and interdisciplinarity, all while keeping a nursing audience in mind and sharing a sense of Philly pride.

Graduate Student Spotlight by Temple University Center for the Advancement of Teaching, 05/2024.

- My unique publication experience during the reflective practicum for the Teaching in Higher Education Certificate was spotlighted on the Center for the Advancement of Teaching web page in Summer 2024.

Graduate Student Feature by Temple University Graduate School, 02/2024.

- 1 of 12 graduate students at Temple University to be featured on the Graduate School web page and social media in Spring 2024. It was a great opportunity to share my research and show gratitude for the opportunities I got at Temple.

CODING LANGUAGES

python, R, MATLAB, bash, JavaScript, Go-lang

REFERENCES

Dr. Sergei L. Kosakovsky Pond, Department of Biology, Temple University, spond@temple.edu

Dr. Stephen MacNeil, Dept. of Computer and Info. Sciences, Temple University, stephen.macneil@temple.edu

Dr. Rob Kulathinal, Department of Biology, Temple University, robkulathinal@temple.edu

Dr. Cihangir Duy, Cancer Epigenetics Institute, Fox Chase Cancer Center, Cihangir.Duy@fccc.edu

Dr. Sayaka Miura, Department of Biology, University of Mississippi, smiura@olemiss.edu